

RANJAN SARKAR

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PhD Research Scholar
Department of Artificial Intelligence
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About

I am a PhD scholar at Department of Artificial Intelligence (AI), IIT Kharagpur. I am working in the [RHPI \(Receding Horizon and Predictive Intelligence\)](#) Lab under the supervision of [Dr. Prabhat Kumar Mishra](#). My research focuses on developing new robust RL algorithms with applications in LLM, Robotics, etc.

Research Interest

Machine Learning • Optimization • Reinforcement Learning • State-Space Model

Work Experience

2023 – 2024: Senior Business Analyst • **EXL Service**
Gurgaon, India

- Worked on **BAU** operations.
- Responsible for data pipelining and analytics with **SQL** and **SAS**.

Education

2025 – Present: Indian Institute of Technology Kharagpur
PhD, Artificial Intelligence
CGPA: 9.33/10

2021 – 2023: Indian Institute of Technology Kharagpur
Master of Science (MSc), Mathematics
CGPA: 8.76/10

2018 – 2021: Ramakrishna Mission Vidyamandira (University of Calcutta)
Bachelor of Science (BSc), Mathematics
CGPA: 9.15/10

2018: Raiganj Coronation High School (HS) • WBCHSE
12th Standard, Science Stream
Percentage: 88.6%

2016: Raiganj Coronation High School (HS) • WBBSE
10th Standard
Percentage: 94.57%

Publications

Ranjan Sarkar, Prabhat K. Mishra, “LQG Solution for POMDP without estimating states: A minimum variance approach”, Indian Control Conference (ICC), 2027.

Teaching Assistant

AI31201 • Reinforcement Learning (Autumn 2026)

AI60209 • AI for Robot Autonomy (Autumn 2026)

AI61006 • AI for Cyber Physical Systems (Spring 2026)

Course Instructor - Dr. Prabhat Kumar Mishra

AI60211 • Algorithmic and Mathematical Foundations for AI (Autumn 2025)

Course Instructor - Prof. Adway Mitra

Projects

Master's Thesis: Image Compression (2022-2023)

Supervisor: Prof. Swanand Khare, Department of Mathematics, IIT Kharagpur

- **Tensor CP decomposition** is implemented in a Python notebook.
- The **Alternative Least Square (ALS)** Optimization Method has been applied to get the low-rank approximation of the RGB image tensor.

Courses & Skills

- | | |
|----------------------------|---------------------------------|
| • Linear Algebra | • Advanced Numerical Techniques |
| • Probability & Statistics | • Convex Optimization |
| • Real Analysis | • Non-Linear Programming |
| • Machine Learning | • AI for Cyber Physical Systems |
| • Deep Learning | • AI for Robot Autonomy |
| • Artificial Intelligence | |

C, Python, LaTeX, Numpy, Pandas, Sympy, CasADi, PyTorch, TensorFlow, Scikit-Learn, Jupyter, etc.

Awards & Achievements

1. Qualified **Joint CSIR-UGC NET** (June 2024) with **All India Rank (AIR) 83**.
2. **INSPIRE Scholarship** Recipient (2018-2023), by DST (Department of Science and Technology), Govt. of India - awarded for top 1% in WB government HS board examination (2018).

Extra-curricular activities

Photography, Singing, Swimming, Table tennis.